

LE NA HOANG

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Detail-oriented Mechanical Engineering student with a 4.0 GPA (Graduating in December 2027) specializing in robotics and autonomous systems. Experienced in *control systems*, *CAD design*, *HVAC*, *programming*, and *simulation* through research and engineering projects.

EDUCATION

Embry-Riddle Aeronautical University (ERAU)

B.S. Mechanical Engineering

Area of Concentration: Robotics and Autonomous Systems

Daytona Beach, Florida

December 2027

GPA: 4.0

SKILLS

Computer Programming: MATLAB, Python, C++, Arduino

Modeling: SolidWorks, CATIA V5, Revit, ANSYS

Manufacturing & Tools: CNC Machining, Welding, Soldering, Precision Tolerancing, Technical Drawing, FEA, DFA/DFM

PROFESSIONAL EXPERIENCES

HDR Inc., Pennington, New Jersey

May 2026 - August 2026

Mechanical Engineer Intern

- Contributed to HVAC design for the USCGA STEM West Project using Revit to develop ductwork routes, diffuser layouts, coordinate air distribution, and integrate mechanical systems within the building model.
- Used DuctSizer to size ductwork based on required airflow and design parameters, and selected Price square ceiling and laboratory exhaust diffusers for different space functions.
- Used Bluebeam Revu to review mechanical drawings, add design markups, compare revisions, and communicate coordination issues across project disciplines.

ERAU Academic Advanced Center, Daytona Beach, Florida

August 2026 – Present

Engineering Sciences Tutor

- Delivered individualized and group instruction in Statics/Solid Mechanics/Dynamics/Engineering Drawings to support student learning and academic success.
- Mentored students in analyzing forces, moments, stresses, and structural systems using engineering fundamentals and problem-solving techniques.
- Fostered critical thinking and technical competency by helping students tackle challenging engineering mechanics problems and improve course performance.

ENGINEERING PROJECTS

Yarkovsky-Kessler Mitigation Research - Undergraduate Researcher

Aug 2025 – August 2026

- Developed Python-based orbital simulation models to analyze Yarkovsky-induced perturbations affecting lunar debris.
- Modeled debris geometry, thermal dynamics, and orbital decay behavior in simulated space environments.
- Identified that irregularly shaped debris exhibits $\sim 3\times$ faster orbital decay than spherical bodies.

Mandrel Centering Guide Design - CNC Control Student Assistant

Jan 2026 – May 2026

- Assisted ERAU Manufacturing Lab in redesigning a medical-grade mandrel centering guide in SolidWorks for B. Braun Medical's Class B cleanroom manufacturing environment.
- Fabricated a precision prototype using CNC machining, welding, and soldering while maintaining tight dimensional tolerances.
- Evaluated anodized 6061 aluminum for fatigue resistance and cleanroom compatibility in medical device production.

The CATIA Final Project - Desk Lamp: Individual Assignment

Feb 2025 – Apr 2025

- Designed and modeled a multifunctional desk lamp in CATIA, integrating a drawer, phone cradle, and storage compartment.
- Produced fully dimensioned part and assembly drawings to industry standards with tight tolerances and manufacturable geometry.

HONORS AND AWARDS

Dean's List (Fall 2024- Present), Society of Women in Engineering Scholarship (2026), AFCEA STEM Scholarship (2026), Charles River All-Access Scholarship (2026), National STEM Excellence Scholarship (2025); ASME Scholarships (2025-2027), Women in STEM Scholarship (2024-2028), ERAU Excellence Award (2024-2028)